

Applied NLP

DATA 641

Lecture 12 — Applications and Taxonomy of Chatbots

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

History of Chatbots

The history of chatbots and of artificial intelligence in general are pretty intertwined.

- **1950s and '60s:** During this period, computer scientists like Alan Turing and Joseph Weizenbaum laid the foundation for chatbots and AI communication. Alan Turing introduced the concept of the Turing Test, which gauges a machine's ability to exhibit human-like intelligence in conversation. Joseph Weizenbaum's work paved the way for the development of the first chatbot.
- **1966 - Eliza:** Joseph Weizenbaum's creation of Eliza marked a significant milestone in chatbot history. Eliza, often referred to as the "Rogerian Psychotherapist," used simple rule-based techniques and regular expressions to simulate human conversation. Despite users knowing they were interacting with a computer program, they formed emotional connections due to the program's empathetic responses.

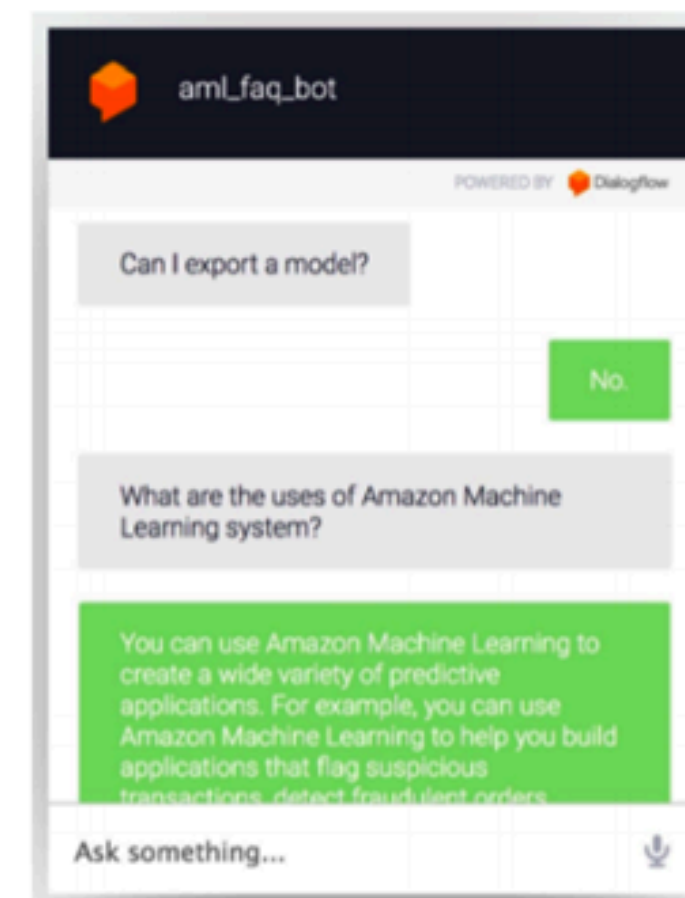
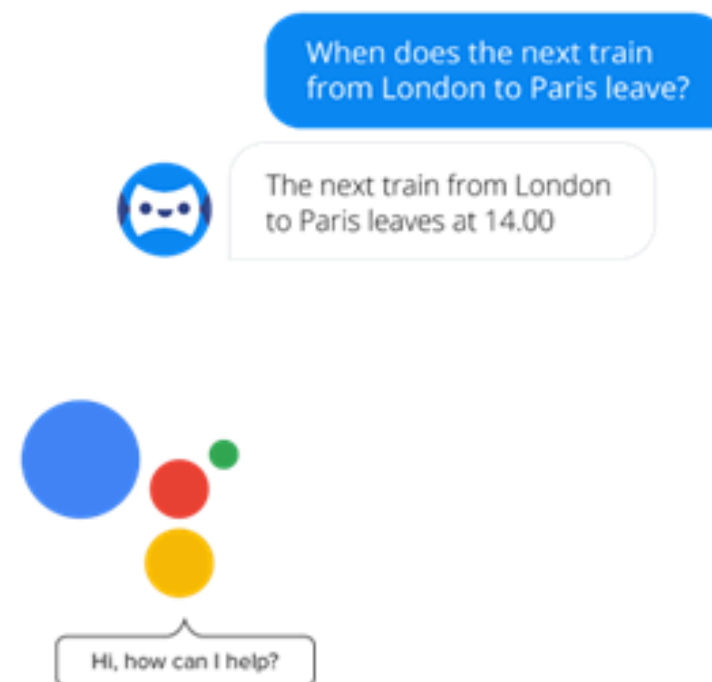
History of Chatbots

- **1980s - 2000s:** During this period, advancements in signal processing tools led to the development of spoken dialog systems. Many of these systems initially had military applications, driven by organizations like DARPA. The primary aim was to improve automated communication with soldiers, enhancing their ability to interact with technology in various scenarios.
- **Modern Era:** From the early 2000s to the present day, chatbots have seen exponential growth in various industries. Natural language processing (NLP) and machine learning techniques have significantly improved their conversational abilities. They are now used extensively in customer service, healthcare, finance, and various other sectors to automate tasks, answer inquiries, and provide personalized experiences.

Dialog systems

Generally, chatbots interact with text but can also use speech interfaces. Examples include...

- Meta messenger
- Google Assistant
- Amazon Alexa



Applications

Chatbots can be used for many different tasks in many different industries, from retail, to news, and even the medical field.

- **Shopping and e-commerce:** Recently, chatbots are being used for various e-commerce operations, including placing or modifying an order, payment, etc. Bots for recommending various items are also of great interest to the e-commerce industry. Industries are focused on building conversational recommendation systems to provide a more seamless user experience.
- **News and content discovery:** Similar to e-commerce, chatbots can be used in news and content discovery. Users may specify various nuances of their search in a conversational manner, and the bot should be able respond with relevant articles.
- **Customer service:** Customer service is another area where bots are used heavily. They are used to lodge complaints, help answer FAQs, and navigate queries in pre-defined conversational flows set by the business requirements.

Applications

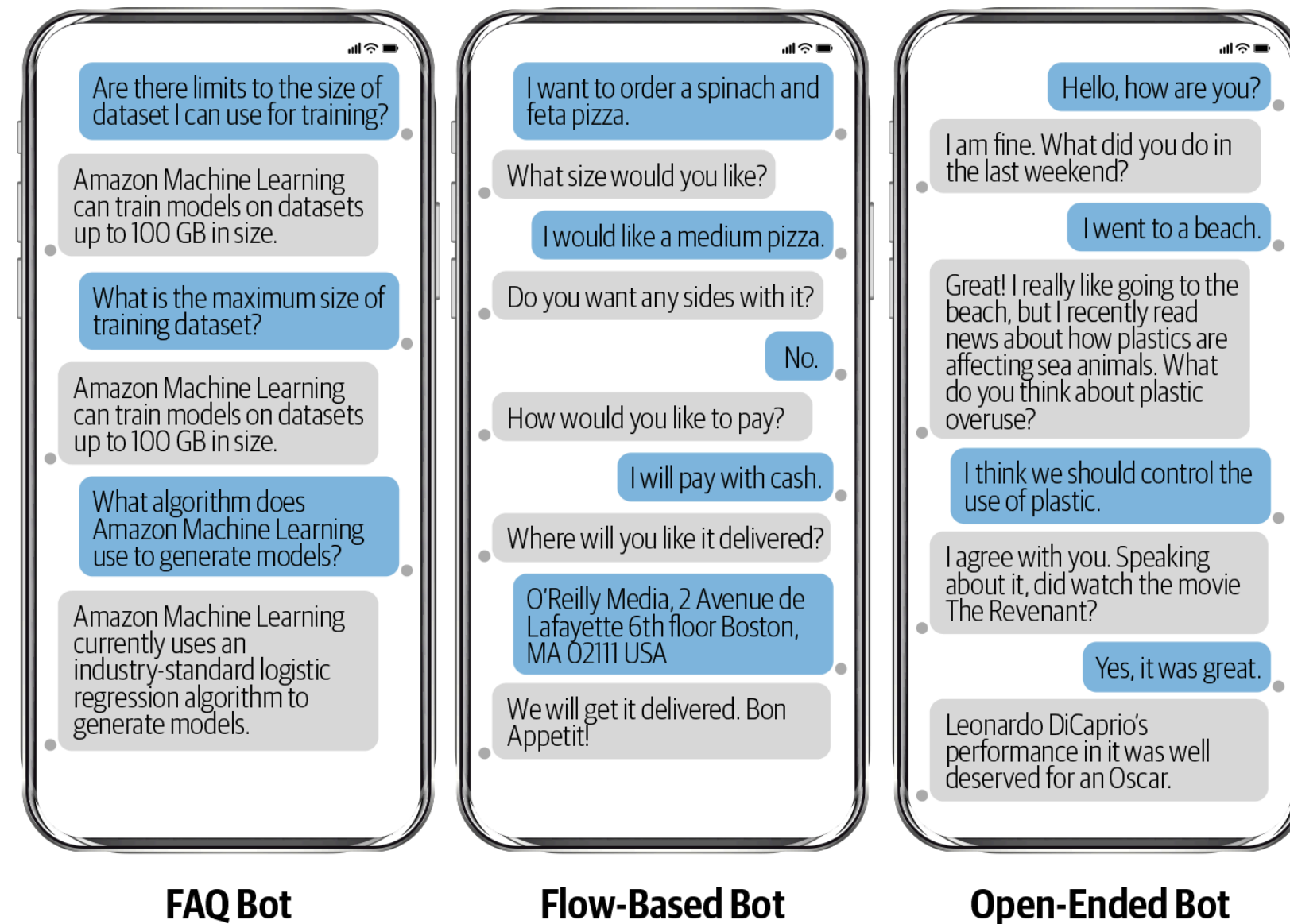
- **Medical:** In health and medical applications, FAQ bots are of great use. These bots can help patients fetch relevant information quickly based on their symptoms. Recently, there has also been interest in building chatbots that elicit useful information from patients, especially older patients, regarding their health conditions by asking relevant questions.
- **Legal:** In legal applications, bots can also be used to serve FAQs for users. They can even be used for more complex goals, such as asking follow-up questions. For example, if a user asks for legal articles to follow up on a case, a bot might ask specific questions regarding the nature of the case to find a more appropriate match.
- **Education:** Chatbots can play a significant role in education. They can be used as virtual tutors, providing students with instant answers to questions related to their coursework. Chatbots can also assist with language learning, helping users practice conversational skills or providing grammar and vocabulary assistance. Additionally, in educational institutions, chatbots can handle administrative tasks such as course registration, scheduling, and answering common inquiries from students.

A Taxonomy of Chatbots

Let us expand on chatbots of various uses and their applicability to various domains. Chatbots can be classified in many ways, which affects how they are built and where they are used.

- **Exact answer or FAQ bot with limited conversations:** These chatbots are linked to a fixed set of responses and retrieve a correct response based on understanding the user's query. For example, if we build an FAQ bot, the bot has to understand the question and retrieve a fixed, correct answer for it. Generally, one response from the user does not depend on the previous responses.
- **Flow-based bot:** Flow-based conversational bots are generally more complex than FAQ bots in terms of the variability of their responses. Users may gradually express their opinions or requests over the course of conversations. For example, when ordering a pizza, a user may express their requested toppings, pizza size, and other nuances gradually. The bot should understand and track this information throughout the conversation to successfully generate a response every time.
- **Open-ended bot:** Open-ended bots are intended mainly for entertainment, where the bot is supposed to converse with the user about various topics. The bot does not have to maintain specific directions or flows of the conversation.

A Taxonomy of Chatbots



Chatbots are classified into two broad categories: (1) goal-oriented dialogs and (2) chitchats. FAQ bots and flow-based bots fall into the first category, whereas open-ended bots are mainly chitchat types. Both of these types of bots are used heavily in industry and are also in the active area of research in academia.